An Educational Human Digital Twin Proposed Model for Personalized E-Learning

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***Abstract*— Global education systems are shifting from traditional one-size-fits-all learning approaches towards a more personalized, student-centered approach so that students can progress at their own pace. Several personalized learning techniques were used, however, these techniques depended mainly on historical data. This research proposes a framework for real-time based personalized learning system. The system depends on real time data and digital twins to provide a more effective personalized learning experience for higher education students. Depending on real-time data about students is expected to provide a more effective personalized learning experience. In addition, it is expected to make the personalized learning process more adaptive. This research states the results of phase 1 of a financed project that aims to build a prototype of a student digital twin for personalized learning. The project is also extended to the development of academic staff members' knowledge, skills and study material improvements. However, this research discusses these points briefly and focuses on the discussion of students personalized learning. In this research the appropriate attributes for building a student digital twin, and the suitable tools for capturing the real-time data of students are determined. In addition, the architecture of the digital twin based personalized learning system, the real-time personalized learning process, the potential insights and decisions are discussed. As for future work, the project continues to the modelling of algorithms for the student digital twin and actual building of the digital twin. It is then extended to setting and using the tools and testing the results.**

Keywords: Digital Twin, Human Digital twin, Personalized learning, personalized training, adaptive learning

I. INTRODUCTION

In modern education, e-learning is getting higher importance as it can be more effective in several ways. For example, it was found that students become able to learn faster online. Li and Lalani (2020) mentioned that research shows that during learning about 40-60% less time is required to learn online than in a physical environment because students can learn at their own pace, scroll backward and forward through the material as needed, and skip, or accelerate through concepts as required. In addition, online students retain 25-60% more material compared to only 8-10% in a physical learning environment.

Due to the importance of online education, a new hybrid model of education has emerged and online education is becoming an integral component. However, online education by itself is not enough. It is needed to meet the needs of modern students that differ much from old students' needs. Today's students face challenges such as work commitments, family obligations, financial constraints… etc. Therefore, educational systems all over the globe are shifting from traditional learning approaches in which one-size-fits-all processes and techniques are used into a more personalized, student-centered approach so that students can progress according to their abilities (Bhutoria, 2022). This research proposes a model that makes use of real time data and digital twins to provide a more effective personalized e-learning experience for higher education students. Higher education is selected as the effectiveness of online learning increases as age increases. Thus, e-learning effectiveness increases in higher education than in lower stages (Li & Lalani, 2020).

Digital twin is an emerging and vital technology for digital transformation and intelligent upgrade. It is a virtual representation of a real object or system which is updated from real-time data through IoT devices. Driven by data and model, the digital twin can perform monitoring, simulation, prediction, optimization and reasoning to help decision-making. Modelling and developing a DT is not an easy process, Tao et al. (2022) illustrated that during the process of digital twin implementation, the digital twin model is one of the most important foundations, because the effectiveness

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and accuracy of digital twin models can directly affect the quality of related services. The term digital twin has been recently extended to humans using the term "human digital twin." This term has been applied in diverse fields, including medicine, sports performance, and manufacturing ergonomics (Shengli, 2021). An essential part of a DT model is the required data and since the main gap in literature is the inconsideration of real-time data in personalized learning, digital twin is suggested for use in this research and is discussed in more details in the coming paragraphs.

Real-time data and DT were not explored as means to support personalized learning online. DT was only applied to provide real-time monitoring of student attendance management (Gomerova et al.,2021). In addition, a personalized training model for teachers was developed by Chaipidech et al. (2022). The importance of the model in developing teachers' technological, pedagogical, and content knowledge was also discussed. However, means for providing personalized training for academic staff members have not been studied enough. Finally, means for incorporating real-time insights about students' comprehension level and data about students' progress to develop course material were not introduced. An additional issue that is still under study is how a human digital twin can be modeled. This research proposed HDTs model for students. The outcomes of this research are expected to fill the gaps discussed in this section.

The main aim of this research is to (1) identify the appropriate attributes and tools that can be used to build a human digital twin for a student, (2) determine how these data attribute can be used to reach beneficial insights and decisions, and (3) propose an educational human digital twin-based system architecture that improves personalized learning. The rest of this paper is designed as follows; section 2 provides a review of the related literature. Section 3 represents the proposed model; section 4 discusses the technology and data requirements for the proposed model, Finally, the conclusion, discussion, and future work are provided in section 5.

II. LITERATURE REVIEW

Personalized learning according to U.S. Department of Education is "instruction in which the pace of learning and the instructional approaches are optimized for the needs, requirements and interests of each learner. Learning objectives, instructional approaches, and instructional content may all vary based on learner needs." (National Forum on Education Statistics, 2019). Personalized learning relies on the fact that learners have different cognitive power and abilities, therefore the instructions, assessments, and evaluation should be different for each of them.

Personalized learning is characterized by (1) Student control of his learning process, (2) Emphasis on learning rather than assessments or tests, (3) Mastery-based student progress, and (4) Learning anytime, and anywhere. This leads to the following core components that should be in any personalized learning environment: (1) learner profiles, (2) personal learning paths, (3) flexible learning environments, (4) student agency of learning process, and (5) learner centered educational process. To achieve efficient personalized learning, individual differences among learners and their psychological diversity needs to be considered to achieve an effective and productive learning process that is planned by considering these individual differences.

Simek (2012) argued that there are several individual differences that affect performance of learners during the learning process, the most common differences are: gender, age, intelligence, ability, interest, prior knowledge, learning style, and motivation. Within each individual difference learners differ to a great degree, therefore, types within each individual difference are determined. It is important to mention that individual differences are qualitative in nature, thus, learners are either in one subcategory or in another, rather than being in one subcategory to a certain degree and in another to a different degree. Each learner is assigned to a different subcategory based on the results of assessment. This is a very difficult process therefore incorporating DT and AI into the process would make it simpler and more accurate.

The personalized learning process considers the learners' historical and current data on activities and performances. This data is used so that each learner gets his own plan based on his performance, grades, and comprehension level. Personalized learning systems should provide tailored learning resources and tools to help learners to study with respect to their abilities to promote fairness in classrooms. Unlike traditional learning, personalized learning makes it easier for instructors to determine each learner's comprehension of the content by offering continuous information about how each learner is progressing toward learning objective. Table 1 below summarizes the findings from related personalized learning research. The table illustrates the extent to which personalized learning was discussed. The table also classifies the type of data used in the research as historic or real-time and illustrates whether digital twins were included in the research.

Personalized learning models based on AI and digital twins have the potential for deep social impact and can improve the overall quality of both the learning process and teaching process (Furini et al., 2022). The instance of digital twin mostly concentrates on a particular application, which is according to this research education and personalized learning. The main focus of the research is modeling for humans, few modeling for humans' digital twins were made and in limited fields such as healthcare. This is because

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modeling a human for health-related issues is relatively simple as it will mostly depend on some vital signs. However, Human digital twins are still a topic under study.

Human digital twins have the potential to change the practice of human system integration by employing real-time sensing and feedback to measurements of human performance, behavior, and environmental influences. However, this concept is relatively new; the literature lacks inclusive and precise definitions for it. Miller, and Spatz, (2022) reviewed literature on human digital twins to provide a generalized structure of these systems. They suggested that components of human models sufficient to provide robust human digital twins are likely to be derived across multiple fields of study. Thus, development of these systems would benefit an open multi-disciplinary research effort.

Shengli (2021) presented HDT as a digital copy in cyber space of a real person in our physical world. Shengli described an HDT as a person's incarnation in cyber space and essentially a model or database which records his information such as: (1) historic data (age, weight, gender, diseases, relatives etc., (2) timely data, and (3) data from relatives. Information is synchronized through communication between the real person and cyber space via Internet technologies. This data is used for further analysis to provide feedback information such as diagnoses, prediction, or other suggestions to you.

Table 1: literature review summary; related literature focus

Author	Brief	Personalized Learning	Data type Historical(H) Real-Time (R)	Digital Twin
Deeva et al. (2021)	conceptualized Student adaptiveness framework to classify Technologies for feedback considering individual characteristics of the student – either determined through a questionnaire or derived from the logged behavior."	√ Automated Feedback	H	x
Sun et al. (2019)	reviewed the literature on computerized adaptive learning systems and developed a classification framework in which real-time analysis of learning activities was made to encourage /warn students based on their activities.	√ Motivation (encouragement and warning)	R	x
Rodzman et al. (2020)	developed a personalized learning system applying rule-based Machine learning methods to improve the learner's performance in Malaysia.	√ Assessment and feedback	R	x
Melesko & Kurilovas (2018)	presented an AI and semantic clustering-based learning analytics software agent for determining learning styles of learners.	√ Learning style prediction	H	x
Holmes (2018)	developed a system based on AI and image processing techniques to classify comprehension and non-comprehension states in near real-time	x	R	x
Gomerova, et al.(2021)	System that monitors student attendance by capturing pictures of students sitting in front of a computer in the lab and matching it with students' records.	x	R	√
Fourtané, (2022)	Build virtual labs to model things like physical resources or places so that each student can learn and experiment at their own pace and run several simulations to better explore behavior under different conditions and parameters	√ Considered individual pace of study	R	√
Furini et al, (2022)	Introduced the idea of a student DT, however, real-time data are needed for the DT concept to work correctly	√	H	√

HDT, like any DT, consists of the physical entity, the virtual copy and the two-way communication between them. However, surroundings and other entities, the physical space, and virtual surroundings and other Digital Twins should add to the cyber space because environmental factors communication with others are of greater influence on people unlike on machines. An HDT also needs to reflect the mental and biological status of a human as well as people's reactions. Adding these factors will help to describe the Human Digital Twin more comprehensively. Detection of mental activities usually relies on biosensors and brain computer science. People's reactions are both objective and subjective, which can be supported by research on psychology and human behavior. The main component of our proposed digital twin is a student digital twin. A student's digital twin might be created with data related to his academic background and studying activities, his behavior on campus, and his digital educational material consumption (Furini et al., 2022). However, these are basically historical data. To meet the digital twin concept more thoroughly, real-time data is required. A summary of different data sources and different attributes used in personalized learning research is illustrated in table 2. The literature review

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crossed not only related personalized learning research, but also some user experience research to determine suitable humanitarian attributes that can benefit the research scope.

Table 2. literature review summary; the used data

Data / Author	Questio-nnaire	Log activities	Learning activities	Assess-ments	Voice Analysis	Digital Material Consumption	Academic Background	Eye Gaze	Facial Expressions	Mouse Movement/ Clicks
Deeva et al. (2021)	√	√	x	x	x	x	x	x	x	x
Sun et al. (2019)	x	x	√ Real-Time	x	x	x	x	x	x	x
Rodzman et al. (2020)	x	x	x	√ Online	x	x	x	x	x	x
Melesko& Kurilovas (2018)	x	x	√	x	x	x	x	x	x	x
Holmes (2018)	x	x	x	√	x	x	x	x	√ images	x
Furini et al. (2022)	x	x	√	x	x	√	√	x	x	x
A. Bond, et al. (2020)	x	x	x	x	x	x	x	√ Pupil Diameter (cognitive load)	√ Emotions	x
Ahmed & Traore (2007)	x	x	x	x	x	x	x	x	x	√ Hands motor expression vs. mood

Bond et al. (2020) used three psychophysiological metrics to measure the variation in the cognitive and emotional states of user's: cognitive load, emotional valence, and arousal. The participants' pupil diameter is used to measure their cognitive load. The users' emotional valence represents the extent of "happy" facial expressions from which the negative facial expressions are subtracted. An advantage of these technological methods over normal methods, like questionnaires, is that it measures the affective reaction during the interaction and not retrospectively. In addition, it is objective not subjective. Hands movements are also required to assess hand motor expressions that can be then related to mood status as illustrated by Ahmed and Traore (2007).

III. PROPOSED MODEL

This section discusses the framework of the proposed solution, and the system architecture. A basic illustration of the proposed solution, its components, and expected output are represented in figure 1 below. The proposed solution is designed to keep data about students' and professors' status, knowledge, health, and mood. It is intended to extend the functionalities of the proposed solution in the future. The proposed solution is designed to be generic enough that it can be used to improve different educational systems. According to figure 1, data capturing tools such as eye trackers and face readers will be used for gathering real-time data to gather related human attributes e.g. eye gaze, facial expressions...etc. The captured values will be fed as real-time data to the developed prototype for the purpose of analysis and decision making. According to the results from the analysis phase suggestions, guidance, and/or strategies will be provided to students. In addition, in case the analytical results showed high frequency of certain troubles among students regarding a specific course or topic, suggestions for content modification or different learning strategies are suggested to the course professor(s). On the other hand, data about each professor' knowledge, skills, and interests are collected and analyzed to determine each professor's knowledge or skills gaps and propose suitable training sessions in a way that corresponds to their personal interests. An illustration of the proposed real-time personalized learning process is provided in figure 2 below. The process is divided into two main phases, (1) the clustering or profiling phase, and (2) the classification phase. Phase 1 is considered the model learning phase. The individual differences discussed by Simek (2012) were used as guidelines to define different clusters to which students should be classified (student profiling), the age criterion was not

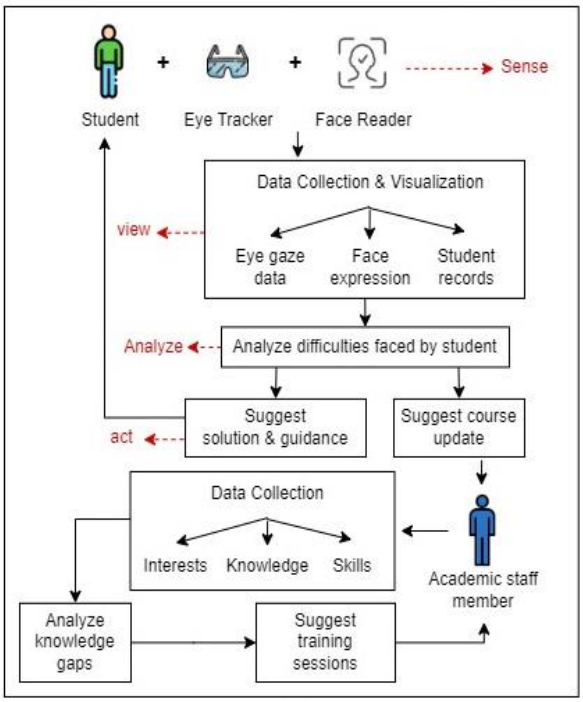
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considered as the research is considering a single age group that of university students. In phase 1, a sample of students is tracked over a certain period. Capturing the student's facial expression, eye movement, eye pupil diameter, voice, text, and participation level should be made repeatedly throughout the specified period. students will be asked to solve an exam, and their scores are kept. The captured data will be matched with the students' grades in the exam to allow the model to learn how the different combination of the extracted parameters reflect on the exam evaluation. The output of this phase is (1) generating different clusters of students, and (2) developing suitable strategies for each cluster. The process of capturing the data will continue for an additional period, thus Phase 2, Prediction by the model, starts. The resulted model from phase 1 will be used to interpret the student performance and a student is classified into one of the generated clusters. Strategies or recommendations are made based on the output.

The objective is to adopt correction activities in case of deviations to put students classified by the model as expected low performance back on track and encourage the strong ones to continue. Example of strategies that can be applied by the system: (1) Send additional material to weak students. (2) Send a survey to weak students to reveal if they suffer specific issues and send survey results to instructor. (3) Invite weak students for online discussions with strong students. (4) Ask the instructor for a recap session. (5) Incorporate gamification strategies (awards, points, achievement badges …etc.). (6) Recommending a resource list to a student who needs help that was effective with successful learners who have similar individual characteristics with the student. The model can also be used to detect the effectiveness of the applied strategy. After a strategy is applied, the newly captured data are analyzed to re-classify the status of the student, i.e., modify his performance level and personal development. If improvements occurred then the strategy is effective and is confirmed as a good solution for this cluster of students, otherwise, it is marked as ineffective for that cluster. This takes us to a higher level, in which adaptive learning is combined with personalized learning, what is known as personalized adaptive learning.

Peng and Spector (2019) defined personalized adaptive learning as "a technology-empowered effective pedagogy which can adaptively adjust teaching strategies timely based on real-time monitored (enabled by smart technology) learners' differences and changes in individual characteristics, individual performance, and personal development." Available machine learning models to analyze emotions from photos, text, and voice are studied and tested for their accuracy. The data layer embeds intelligent strategies to create the student's digital twin and smart approaches to produce enriched digital educational material, starting from on-site lessons and intelligent algorithms to analyze digital twins and to personalize learning experiences.

Figure 1. The Proposed Model



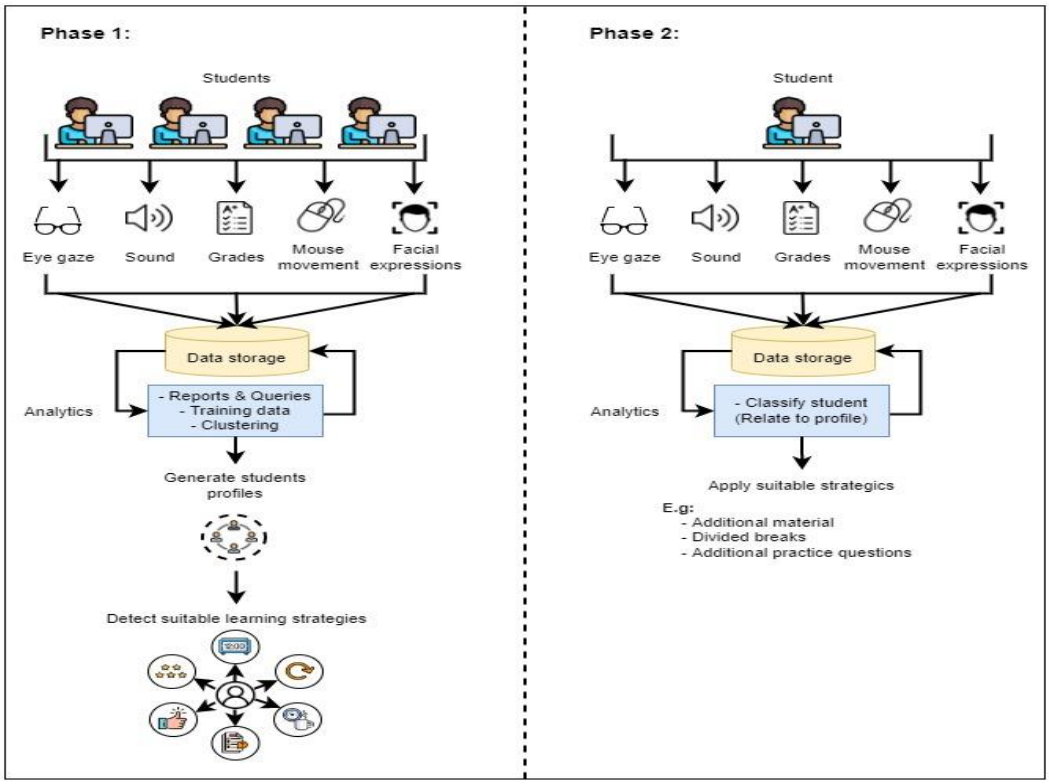
IV. TECHNOLOGY AND DATA REQUIREMENTS

Miller, & Spatz (2022) illustrated that according to the function of human digital twin being developed, an HDT can be modelled using one or more of the following features: 1- Physical (anthropometric attributes, biomechanics attributes, times required for task completion, eye movements, and injuries), 2- Physiological (heart rate, heart rate variability, muscle tension, blood oxygen level, brain electrophysiologic signals, pupillometry, blink rates and timing, peripheral blood flow, and gastronomical activity) which correlate to higher level physiological measures such as fatigue, circadian cycle, and alertness level. 3- Perceptual performance (auditory sensitivity, ability to decipher speech in different languages, visual sensitivity, color sensitivity, contrast sensitivity. 4- Cognitive performance (knowledge, skills, abilities or aptitudes, workload level, situation awareness, decision-making abilities. 5- Personality characteristics (personality type, propensity to trust, and propensity towards suspicion). 6- Emotional state (feelings experienced by the

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individual within their current circumstance, including levels of depression, and anxiety). 7- Ethical stance (representation of the individual's belief system, including values, beliefs, and mores). 8- Behavior (actions taken by the individual to interact with the system).

Figure 2. The proposed personalized learning process



Thus, we conclude building a student digital twin should cover several (personal characteristics, emotions, concentration level...etc.). This complies with Montini et al. (2021) who argued that a human being can be characterized by many different dimensions, each of them requiring a dedicated model. It also complies with Shengli (2021) who argued Human Digital Twin can use a variety of data and information to evaluate the human situation and give evaluation opinions to the real person and what relates to him.

This research proposed facial expressions, eye movements (fixation and saccades are excellent measures of visual attention and interest), eye pupil diameter dilation (which is a measure of high cognitive load given light intensity is constant, voice tone in discussions, EEG signals, mouse movements, together with some personal characteristics can be used as real-time data that reflects level of concentration and comprehension of a student) (Farnsworth, 2018). In addition, computer-mouse movements will be recorded to log-files during the task as literature revealed that a connection between motor expression of the hand and mood exists (Ahmed & Traore, 2007). Moreover, data gathered by student's digital twins in e-learning space can be combined with other data gathered from students in the physical space to have insights about parts that need more clarification, simplification, or practice in course contents. An example of this data could be assessment of the level of knowledge gained by students after the lecture through solving some questions, number of questions raised by students during the lecture, library and classroom time, and evaluation of past exams.

Moreover, the model needs to incorporate student's personal characteristics, performance level, and development level to support the application of personalized adaptive learning, by allowing the learning strategies to adapt to changes in these three factors. The individual characters factor mainly solves the problem related to "what to learn, the individual performance layer addresses "how to learn" problem, and the personal development layer addresses the issue of "how well learned" for their personal vision.

Montini et al (2021) discussed a model for a worker DT, they illustrated that experience, knowledge, skills, capabilities, wellbeing, and performance indicators are all elements that could be included in a worker's digital representation. They also included the surrounding environment i.e.,

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humidity, noise, pollution, or lighting and the items interacting with a worker, like machines, equipment, and robots. Similarly, a student digital twin should include several aspects, those discussed previously: emotions, activity, personal characteristics, and focus. Therefore, a student's relations/interactions with his colleagues, and his surrounding environment should be also considered.

Data capturing tools that can allow for gathering the required data are eye trackers for eye movements, camera, and artificial intelligence tools for eye pupil diameter for focus level assessment, and facial expressions for mood assessment, EEG headsets for cognitive load assessment. The coming section illustrates the resulting benefits of the proposed solution.

V. CONCLUSION, DISCUSSION, AND FUTURE WORK

This research proposes a real-time based personalized learning system that depends on real time data and digital twins to provide a more effective personalized learning experience for higher education students. This research determines the appropriate attributes that need to be used to digitally twin a student, and the suitable tools to capture these attributes. The research also describes how the system works together to achieve its objective. The research is based on the concept of personalized learning, thus, a process for real-time personalized learning was proposed and discussed. Moreover, some learning strategies were highlighted as possible outputs.

This study is filling the literature gap in section two by: (1) identifying attributes that can be used to model and build an HDT for students, (2) making use of real-time data of students instead of historical data to improve their curve of learning and allow students to create their own pace of study, (3) providing directions for course material improvement through guiding professors about areas that majority of students showed low comprehension level or have progressed with low quality in related questions.

The research provides an opportunity for promoting the implementation of Education 4.0 in Egyptian universities, thus, coping with the changes. For higher education institutions, that means figuring out what their future graduates will need. Integrating universities teaching and operations with technological improvements will prepare future graduates for work Education 4.0; a method of learning that is aligned with the fourth industrial revolution.

Other short-term outcomes are assisting professors to better understand each student's potential and limitations and allowing improving the learning experience. Professors can use the system to help them enhance the instructions they provide their students as well as make learning more engaging and exciting. Students can benefit from immediate feedback since it enables them to see where they are doing wrong and how they may improve.

It is important to mention that e-learning is not only tied to education, e-learning can also relate to employees training. Thus, the outcomes of this research can also be applied to allow companies to achieve higher engagement of their employees and better outcome of the training process.

Literature studying personalized learning mostly focuses on developing algorithms that can: (1) predict, or detect students learning style, (2) use students' historical data or quizzes rates to provide reports on their progress, (3) provide tailored learning resources. However, these are mainly restricted to the current knowledge level and learning behavior data, however, real-time data were not considered in the process. Meanwhile, real-time data inclusion in the personalized learning process would allow building a highly efficient study plan for each student. The researcher suggests depending on DT, IoT, and AI in order to capture real-time data, analyze data, and take real-time actions so that the learning process gets more adaptive.

DTs opportunities and potential are huge, it has certainly uncountable possible applications ranging from production to health, from climate change to sustainability, and from management to politics. It is also often suggested that digital twins will allow one to predict the future and implement optimal control. DTs are expanding current personalized services, goods, and devices to the areas of decision making, behavior, and health i.e. HDTs. Potentially, this enables an Internet of Bodies thus allowing one to uniquely identify and track everybody via an e-ID, read out data of body functions, and manipulate body functions, supposedly to improve the health of people. This triggers the need for high security standards to maintain such sensitive data (Helbing, & Sánchez-Vaquerizo, 2023).

There is a need also to stress that modelling humans, their thinking, feeling, decisions, and actions do not only concern the area of brain science, but also the social sciences and humanities, which make it complex quantify and measure related concepts (dignity, creativity, friendship, trust, and love, …etc.). Accordingly, a data-driven society may easily neglect important qualities, which would possibly produce an inhumane society. Treating people just like things or robots would be considered highly inappropriate and unethical. Therefore, open issues for research would be how to monitor human data securely, and model human thinking and feeling the highest possible to prevent the production of inhumane society.

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As for future work, a multidimensional data model for a student will be finalized based on the data discussed previously. Then, the most suitable machine learning models to analyze emotions from photos, text, and voice will be embedded to the algorithm layer of the DT. A prototype of the student DT will be developed according to the model discussed. The prototype of this framework will be applied on a sample of students to ensure it is able to solve the identified problem. Implementation will certainly take place for evaluation purposes. In this regard, working with teams from the university will be inevitable to be able to realize this task. A data set should be used to firstly evaluate the validity of data capturing tools readings to avoid inaccurate interpretation of experimental data. Moreover, before analyzing the data collected to create the digital twin, the error range of the data capturing tools should be eliminated to help remove false variations from readings. Secondly, the evaluation of the prototype in terms of usability and value that it is bringing to the university campus will be done by analyzing the quality of recommendations. Thirdly, evaluation of the acceptance of the proposed prototype will be done.

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